

Certificate of Clinical Validation and Evaluation

Research Title: BanglaDiaHallu: Evaluating Hallucinations in LLMs and the Impact of Guideline-Grounded RAG on Trustworthy Bangla Diabetes Question-Answering

Date of Evaluation: June 5, 2026

Subject of Evaluation:

1. **Questionnaire:** A set of 40 frequently asked Diabetes related questions in the Bangla language, featuring transliterated English medical terminology.
2. **Knowledge Base:** A paragraph-level semantic database derived from the National Guideline on Diabetes Mellitus (2023) and formally recognized supplementary medical literature.
3. **Model Performance:** Responses generated by multiple Large Language Models (LLMs), including the BanglaDiaRAG system.

Face Validity & Question Assessment

We, the undersigned registered medical practitioners, certify that we have curated and clinically reviewed the 40 frequently asked Diabetes-related questions in the Bangla language used in this study. We confirm that these queries represent the most common and clinically significant concerns of diabetic patients and the general population in Bangladesh. The questions are found to be socio-culturally appropriate and medically relevant for a patient-facing educational system.

Knowledge Base & RAG Evaluation

We have evaluated the Bangla-translated knowledge base used for the Retrieval-Augmented Generation (RAG) pipeline. We certify that the retrieval chunks are factually consistent with the **National Guideline on Diabetes Mellitus (2023)** and provide sufficient clinical context for generating safe and accurate medical advice.

Model Output Evaluation

Following a randomized, model-blinded joint evaluation protocol, we operated as a unified clinical consensus panel to assess the generated responses. To ensure absolute data integrity and eliminate individual scoring bias, each model output was reviewed collectively. Any initial variance in interpretation was formally debated and resolved via multi-expert clinical consensus prior to final data entry.

The evaluation was executed across the following five core clinical and linguistic dimensions:

- **Medical Accuracy:** Factual correctness and absolute alignment with the clinical gold standards of the National Guidelines.
- **Completeness:** The breadth and depth of coverage relative to the multi-part concerns raised in the patient's native query.
- **Safety:** The absolute absence of clinically harmful advice and the successful inclusion of necessary medical guardrails and emergency disclaimers.
- **Readability:** The linguistic naturalness, empathetic tone, and socio-cultural appropriateness of the code-mixed Bangla syntax.

- **Reliability:** The structural and semantic consistency of the model's clinical reasoning across multiple discrete generation iterations.

Final Declaration

Based on our joint clinical expertise and the empirical data compiled through this consensus panel evaluation, we formally certify that the BanglaDiaRAG system demonstrates a profound, mathematically and clinically verifiable reduction in critical hallucinations when compared against baseline open-source LLM configurations.

We validate that the RAG architecture successfully transforms lightweight models to deliver safe, highly guideline-adherent, and medically grounded responses tailored to the demographic realities of Bangladesh. The methodologies executed for foundational knowledge base construction, dataset curation, blinded consensus evaluation, and output verification adhere strictly to established international scientific and clinical research standards.

Evaluator 1

Signature: [Signature]
 Name: OMMA HAFSA ANY
 Designation: Professor
 BMDC Reg. No: A 45488
 Institution: Army Medical college Jashore
 05.06.26

Dr. Omma Hafsa Any
 MBBS, M. Phil
 Professor & Head
 Department of Pharmacology & Therapeutics
 Army Medical College Jashore

Evaluator 2

Signature: [Signature]
 Name: K. M. ROCKYBUL HASSAN
 Designation: Assistant Professor
 BMDC Reg. No: A-88110
 Institution: Army Medical college Jashore
 05.06.26

Dr. K. M. Rockybul Hassan
 MBBS (DU); MPhil (BSMMU)
 Assistant Professor
 Department of Pharmacology
 Army Medical College Jashore